AI-compute supply chain: chart pack

Current state of the analysis. Data: UN Comtrade + TDM monthly panel (60 HS6 codes, 2017-01..2026-04, audited vs Atlas) and Atlas HS2012 annual.
Sections: 1 the chain as flows (2024) -- 2 the time-varying factor model -- 3 concentration & fragmentation.
Details: docs/data.md, docs/supply-chain-narrative.md, results/.

2026-07-27

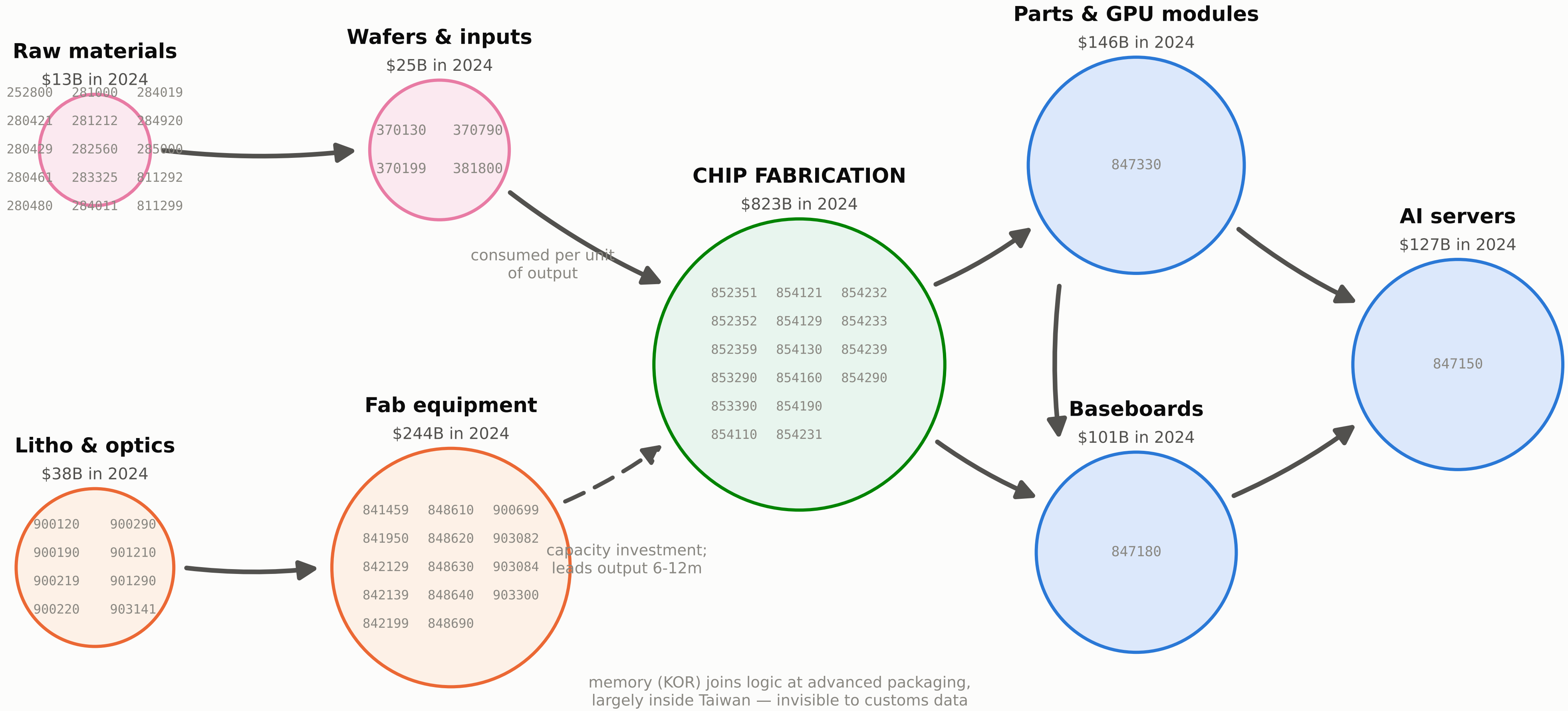
1. The chain as flows

2024 bilateral flows, China+Hong Kong merged (CHK), intra-bloc excluded. Stages 1-5 from Atlas HS2012 annual data; stages 6-8 from the monthly panel (docs/data.md §3). Generated by scripts/export_supply_chain_sankey.py.

The chain's stylized shape, with HS6 codes per node

Two parallel input branches (materials; tools) converge on chip fabrication; the output side is sequential. Node heights scale with the log of 2024 world trade. Solid arrows: consumed per unit of output; dashed: capacity investment (leads output 6-12 months in fab countries).

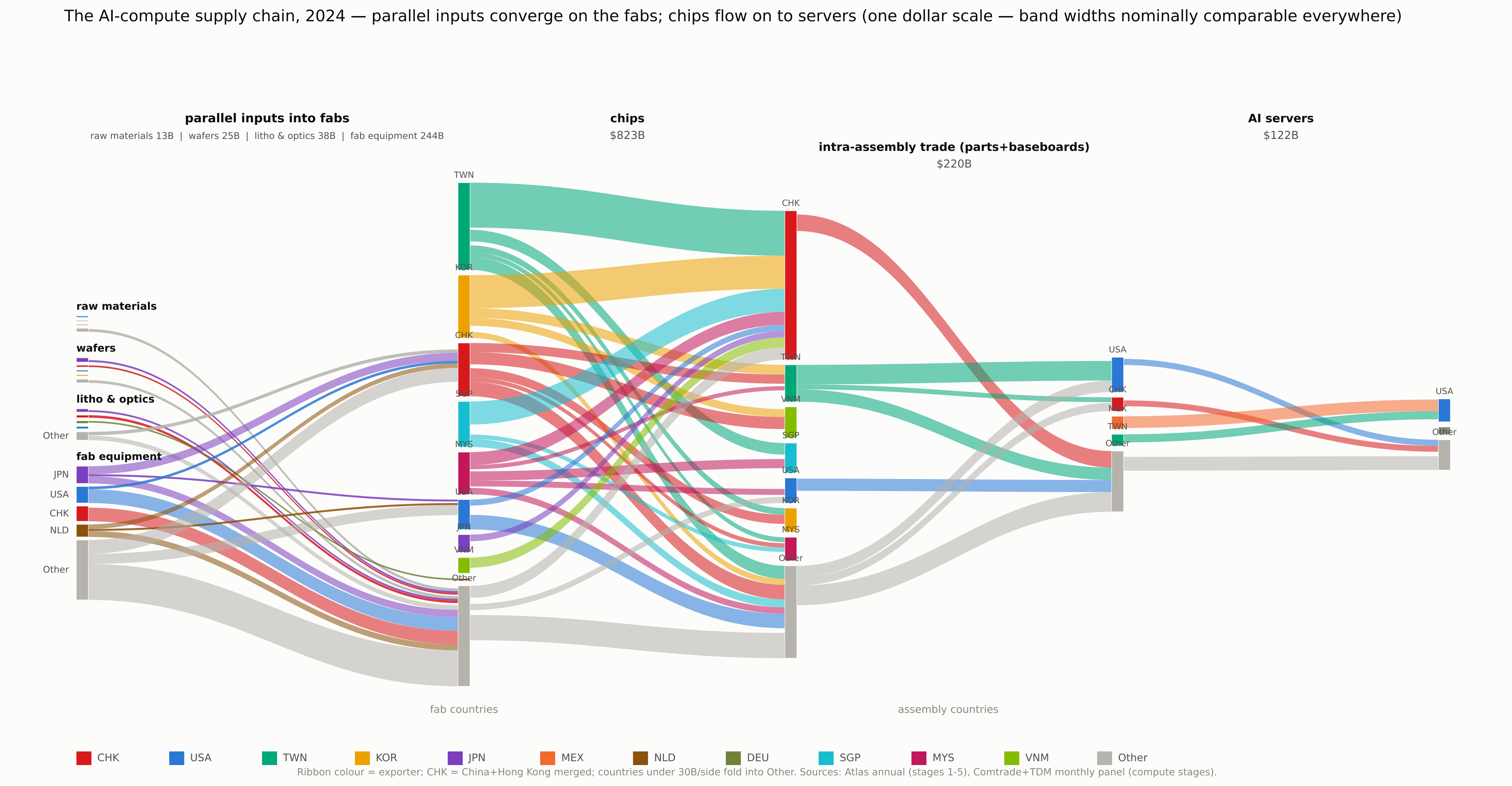
The AI-compute supply chain — stylized shape, with the HS6 codes in each node



Two parallel input branches converge on fabrication; the output side is sequential. Circle diameters follow the log of 2024 world trade (illustrative, so chips does not dwarf raw materials). The materials branch (pink) is consumed per unit of output; the equipment branch (orange) is capacity investment (solid vs dashed arrows mark the same distinction). Design/EDA/IP value enters as services, never as goods trade. The three AI-compute codes of the Fed basket -- the monthly panel's focus -- are highlighted blue. Some small niche inputs sit outside the 60 codes (laser sources, vacuum pumps and valves, ABF substrates -- see notes/firm-level-supply-chain-data.md). Codes: OECD semiconductor mapping + Fed AI-compute basket (docs/data.md §1).

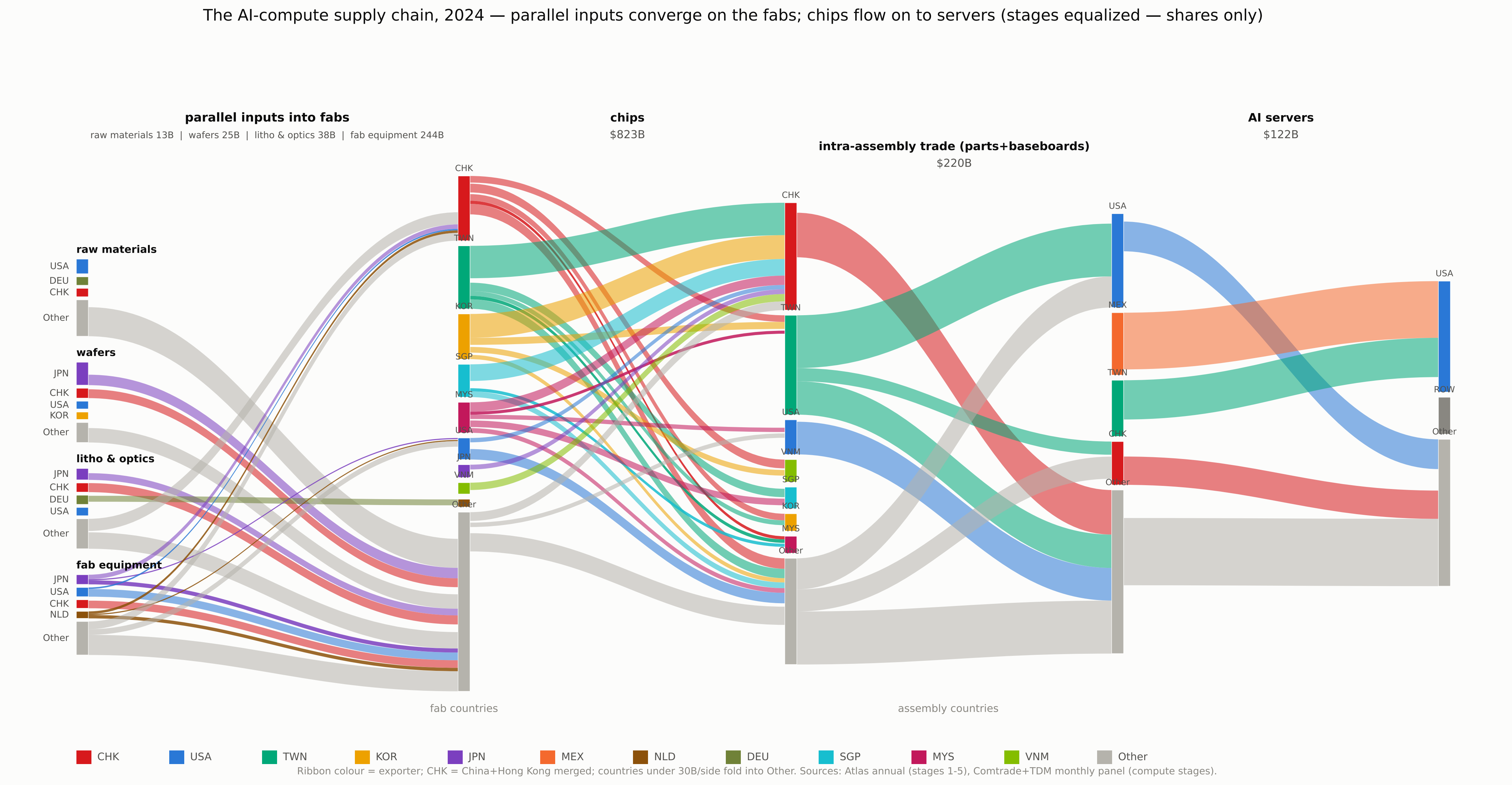
Whole-chain overview, dollar scale (coarse)

All eight stages left to right; band width = 2024 dollars. Inputs, litho optics and equipment run in parallel and converge on the fab countries; parts and baseboards form the intra-assembly segment before servers.



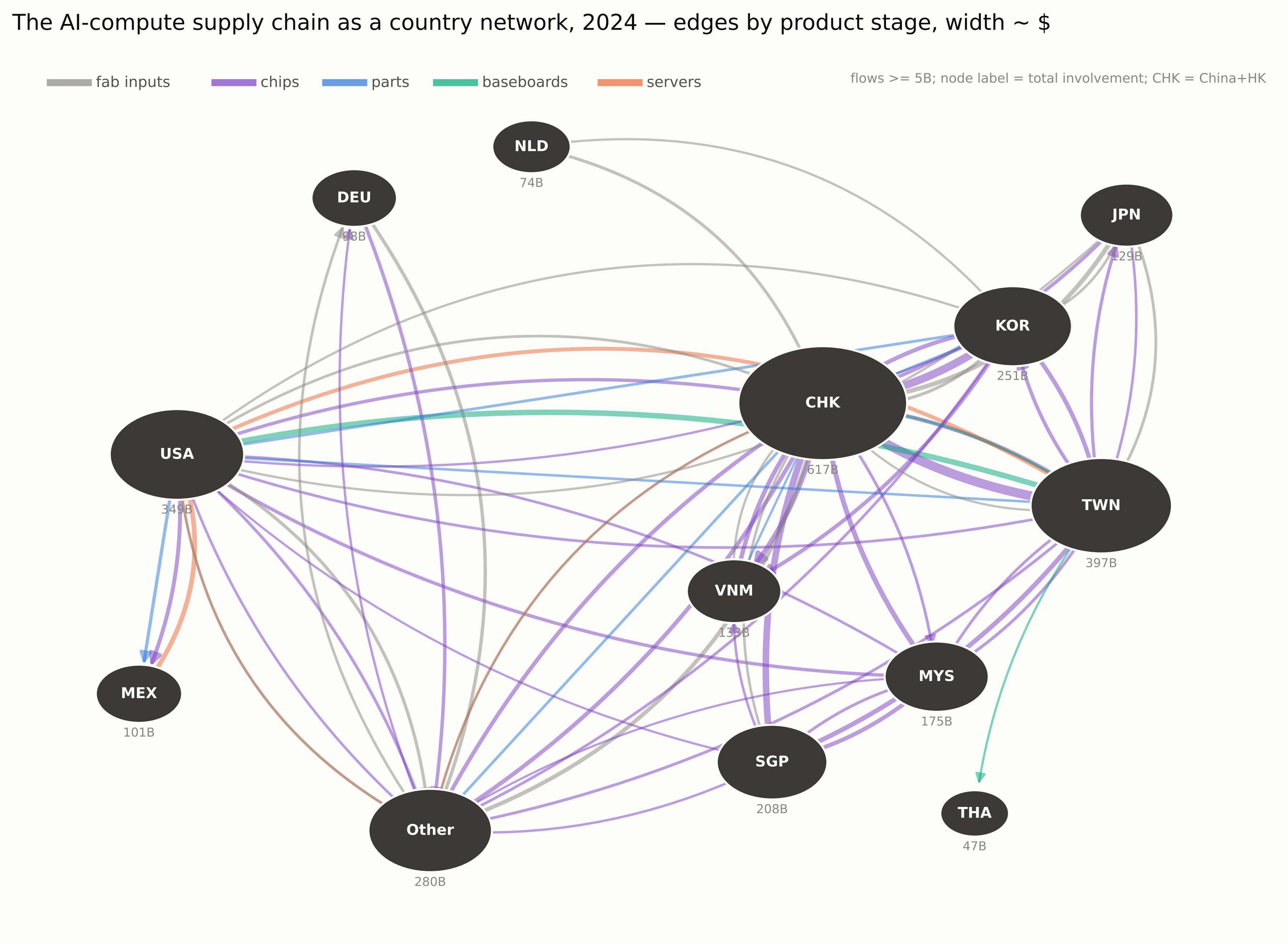
Whole-chain overview, normalized (coarse)

Same flows with each stage rescaled to equal width, so small stages' composition is readable (dollar scale makes chips dwarf everything).



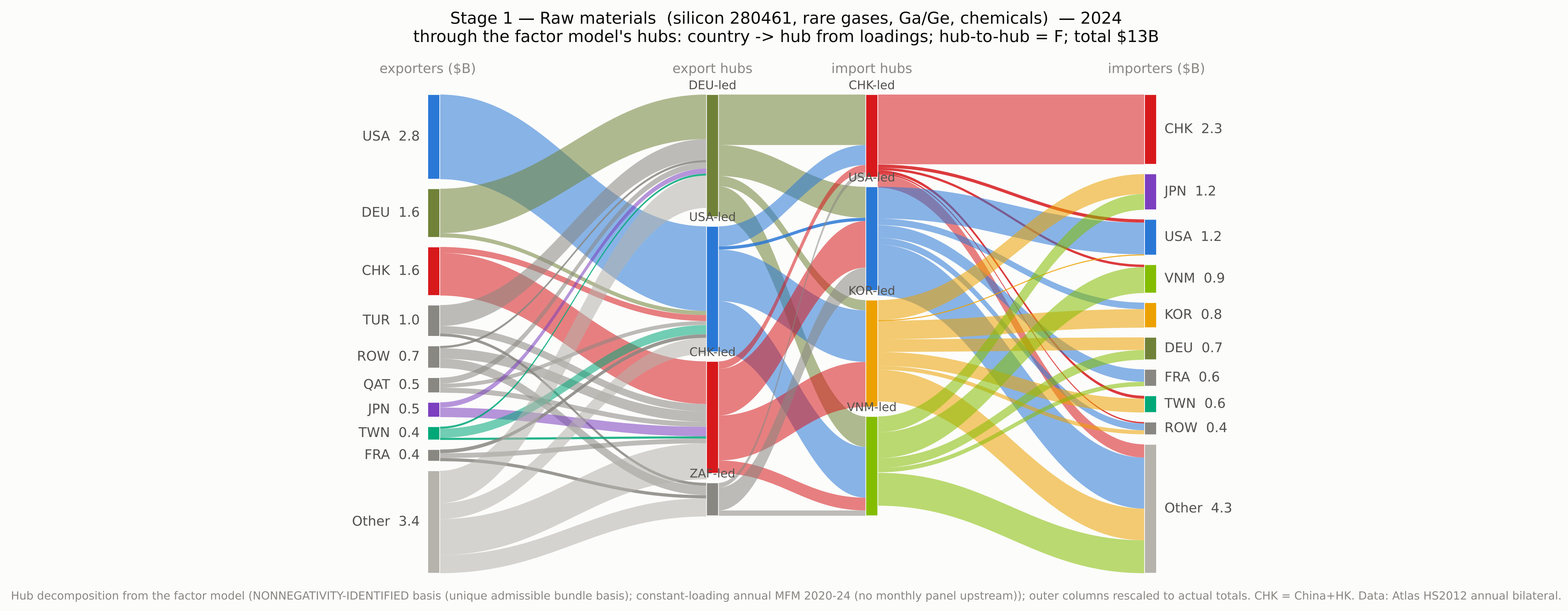
Country network, all stages

Nodes = countries, edges = 2024 flows colored by stage. Shows integrated roles a stage sequence cannot: Mexico takes in chips and parts and ships out servers at a single node.

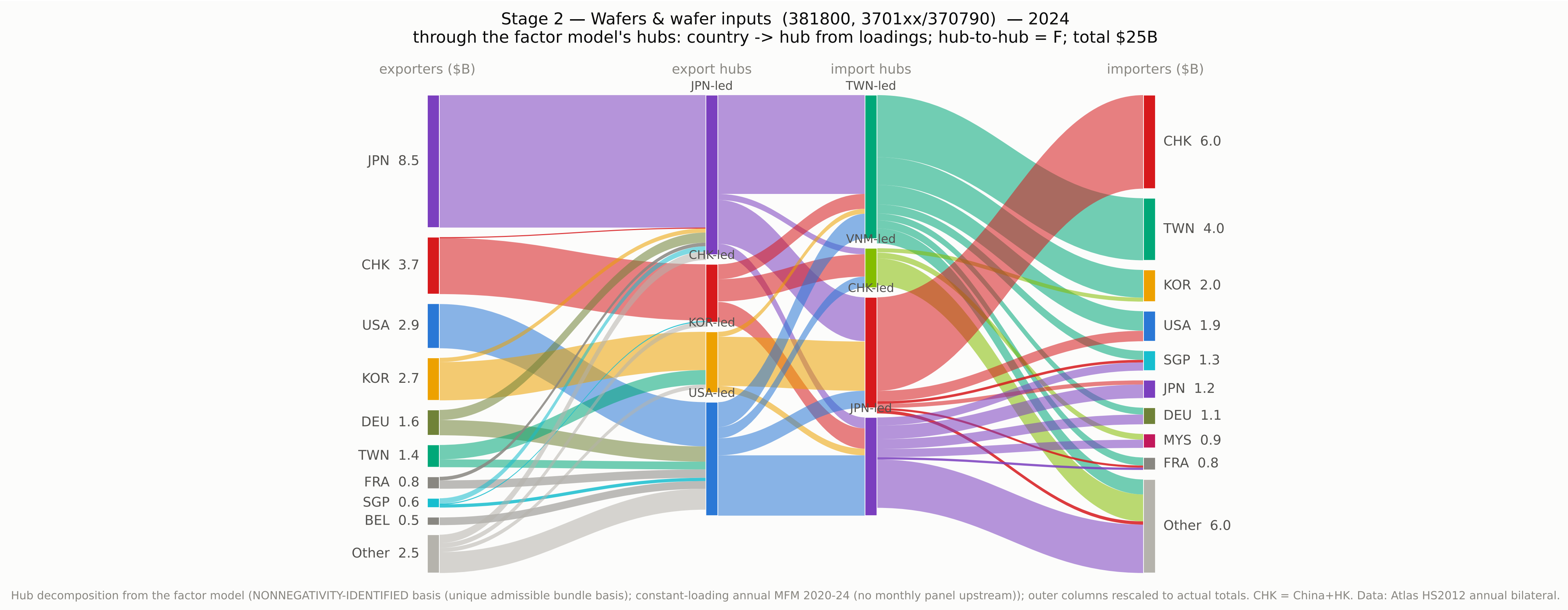


Stage 1 -- raw materials, hub decomposition

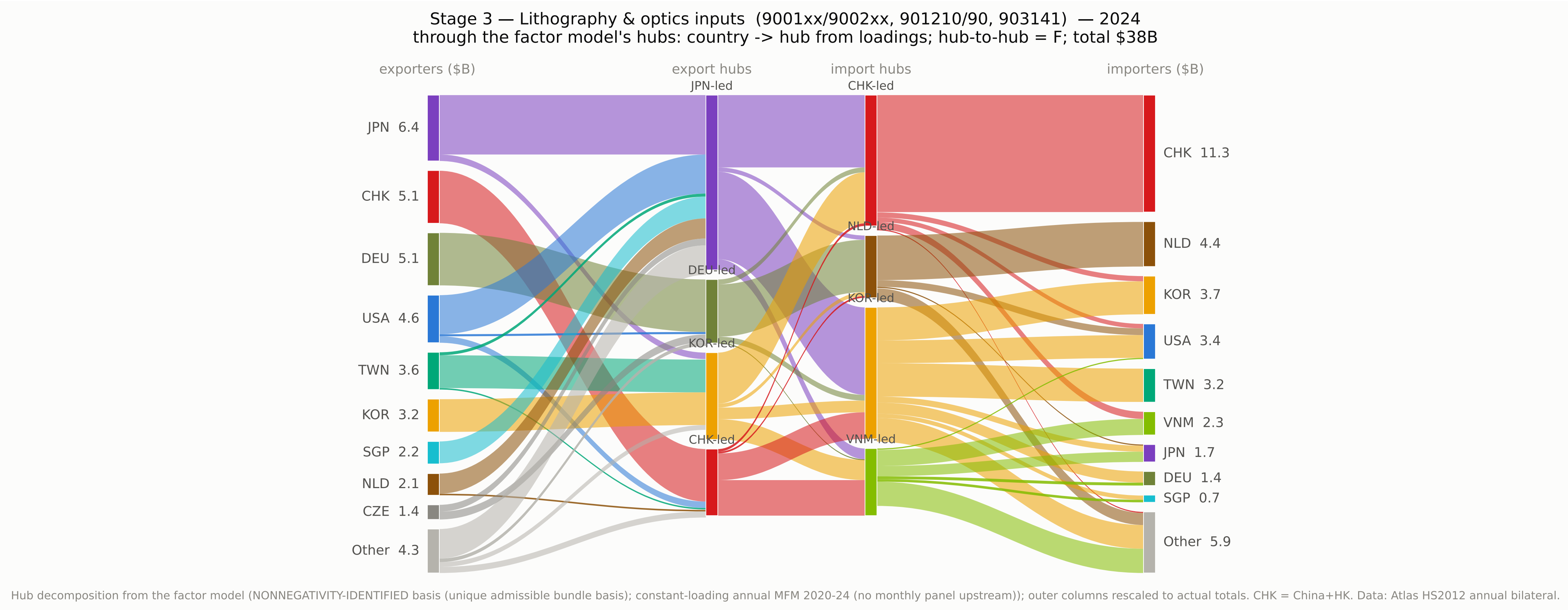
Four-column view: exporters -> export hubs -> import hubs -> importers, hubs estimated by MFM in the nonnegativity-identified basis. Applies to every stage page that follows; fits R^2 0.95-0.99.



Stage 2 -- wafers & wafer inputs

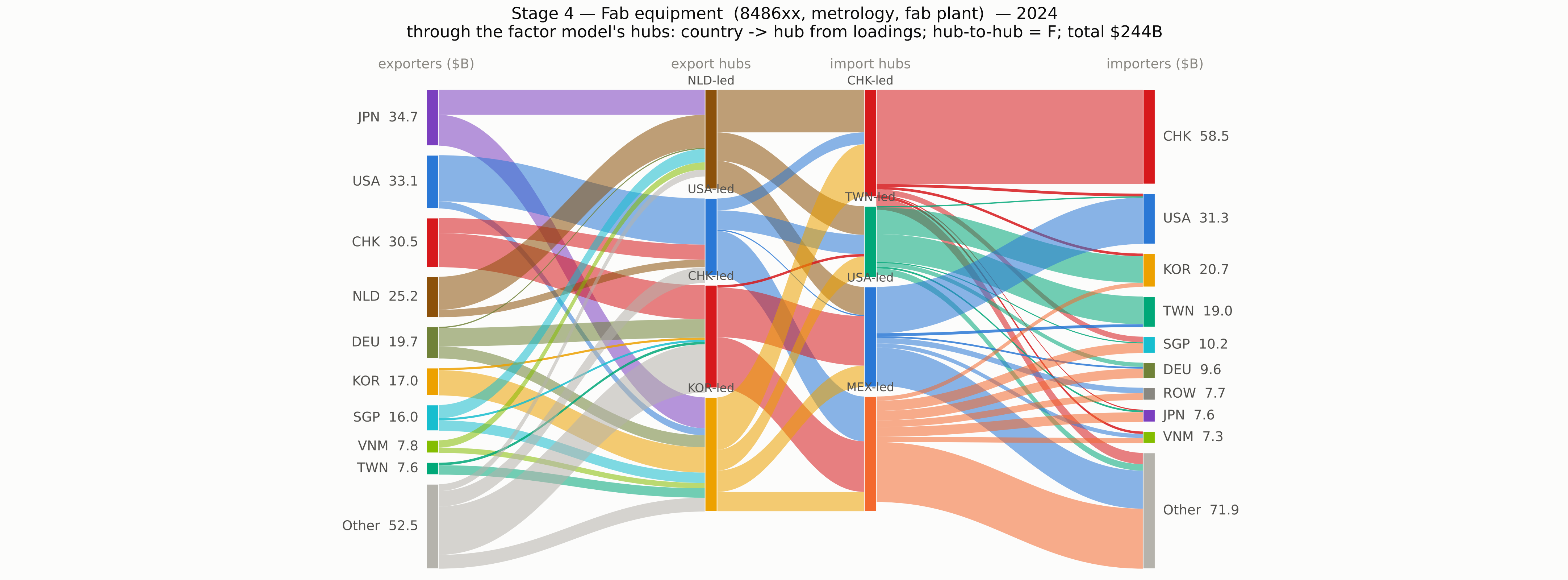


Stage 3 -- lithography & optics inputs



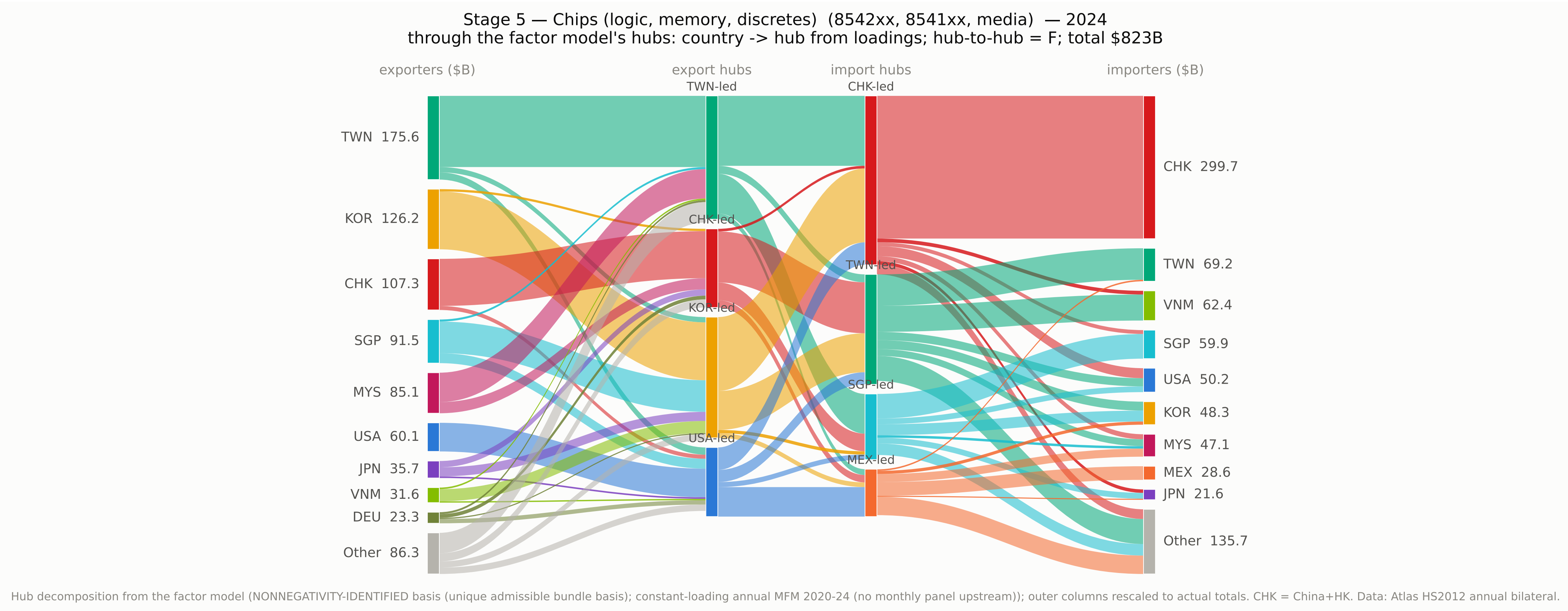
Stage 4 -- fab equipment

Capacity goods, not consumables: in fab countries these imports lead chip exports by 6-12 months (docs/notes/edge-type-tests.md).



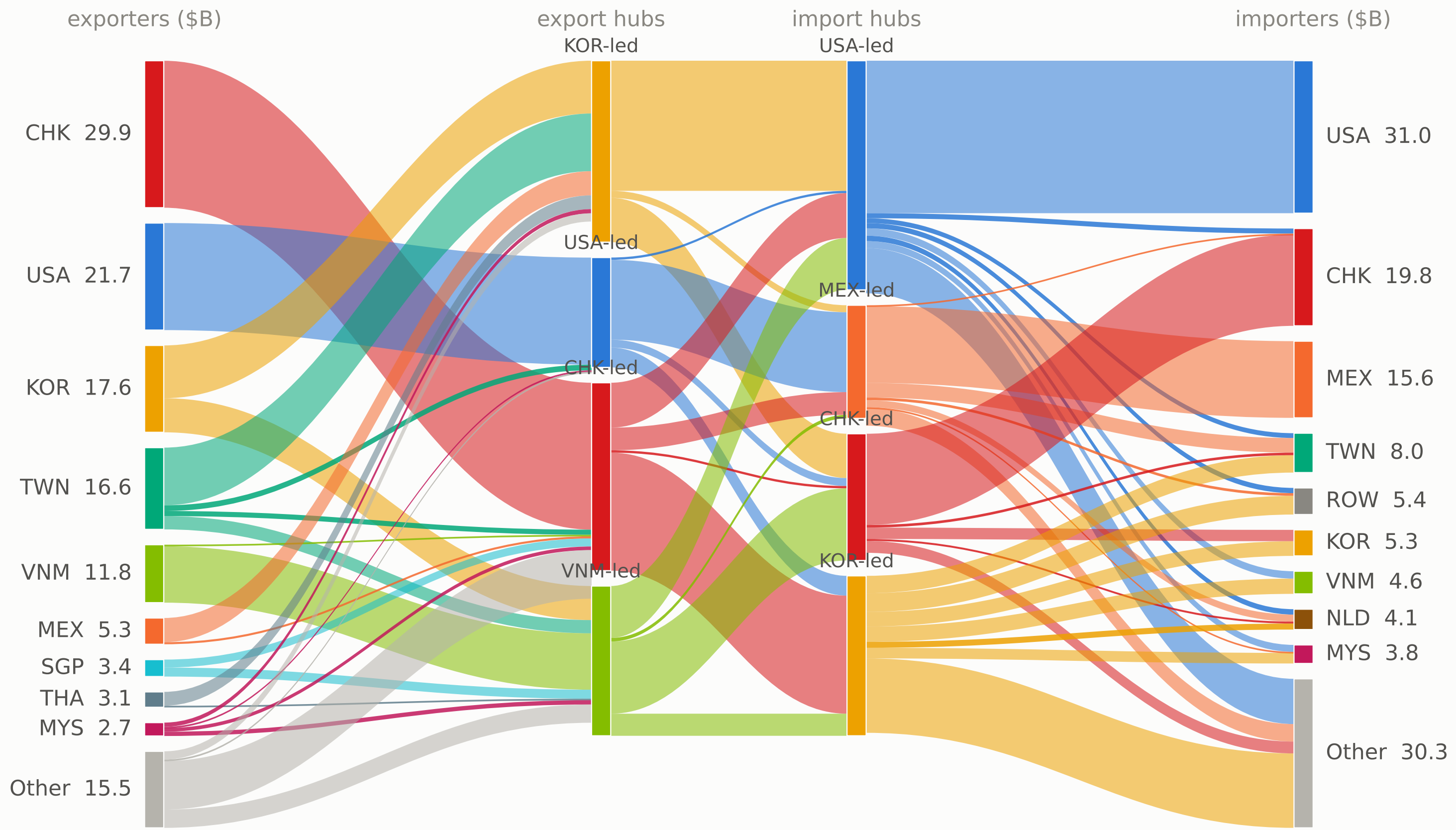
Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 (no monthly panel upstream)); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral.

Stage 5 -- chips (logic, memory, discretes)



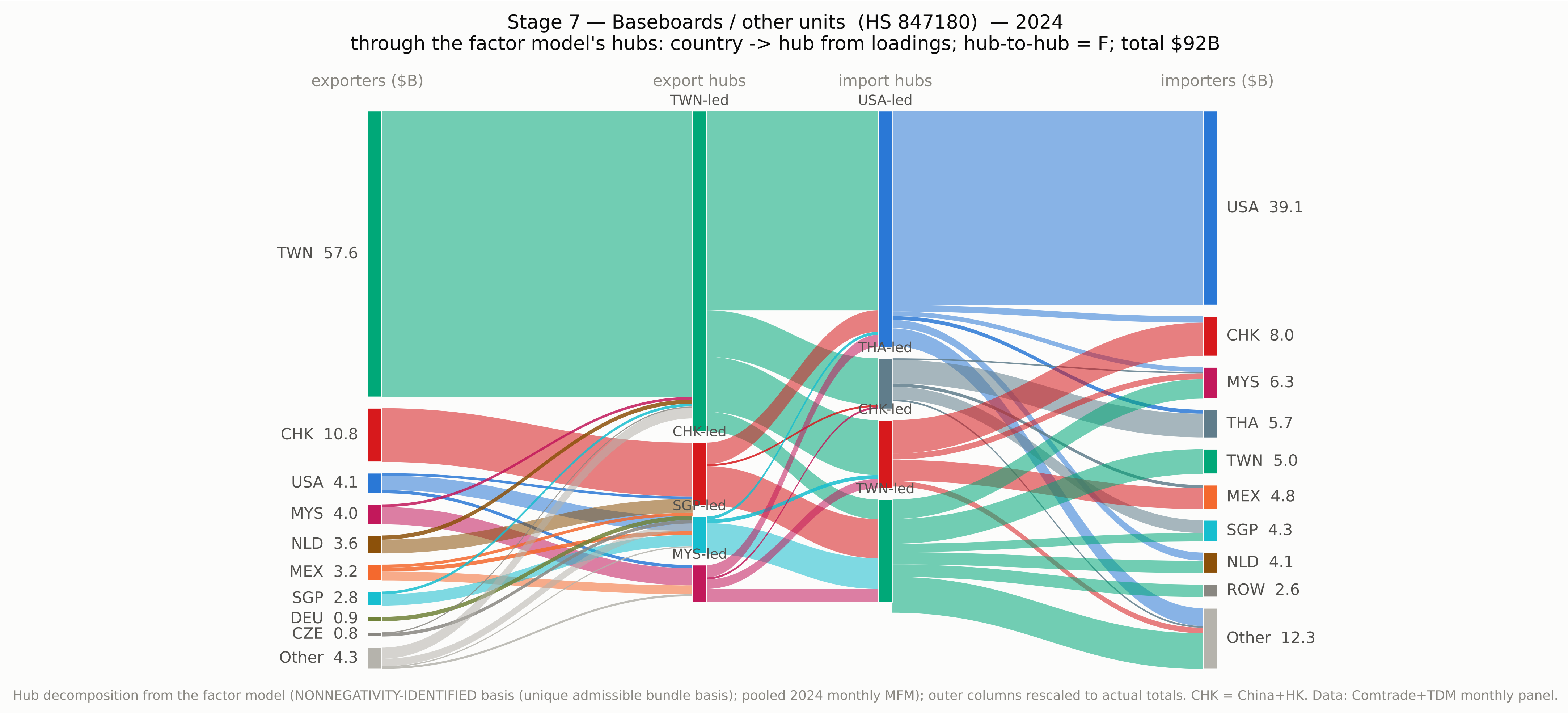
Stage 6 -- parts & GPU modules (847330)

Stage 6 — Parts & GPU modules (HS 847330) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total $128B

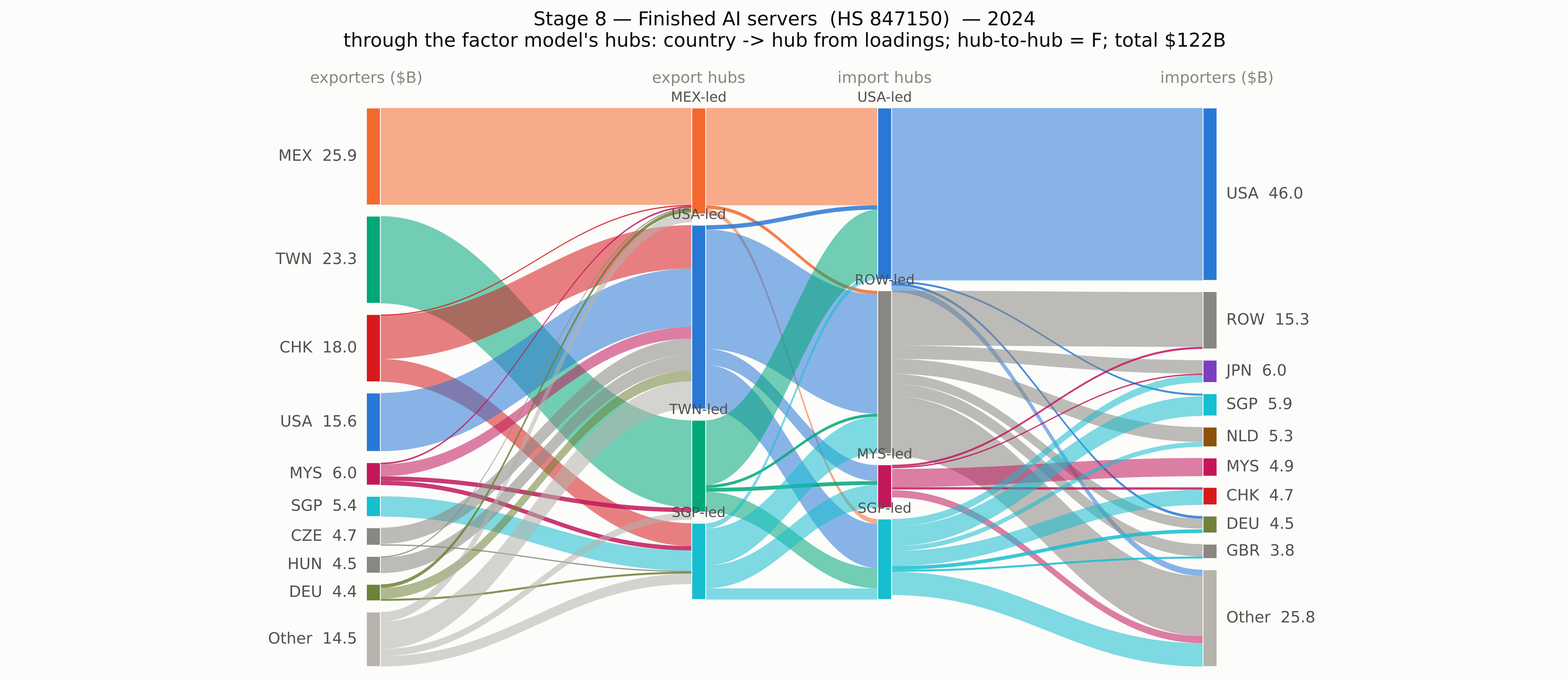


Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); pooled 2024 monthly MFM); outer columns rescaled to actual totals. CHK = China+HK. Data: Comtrade+TDM monthly panel.

Stage 7 -- baseboards (847180)



Stage 8 -- AI servers (847150)



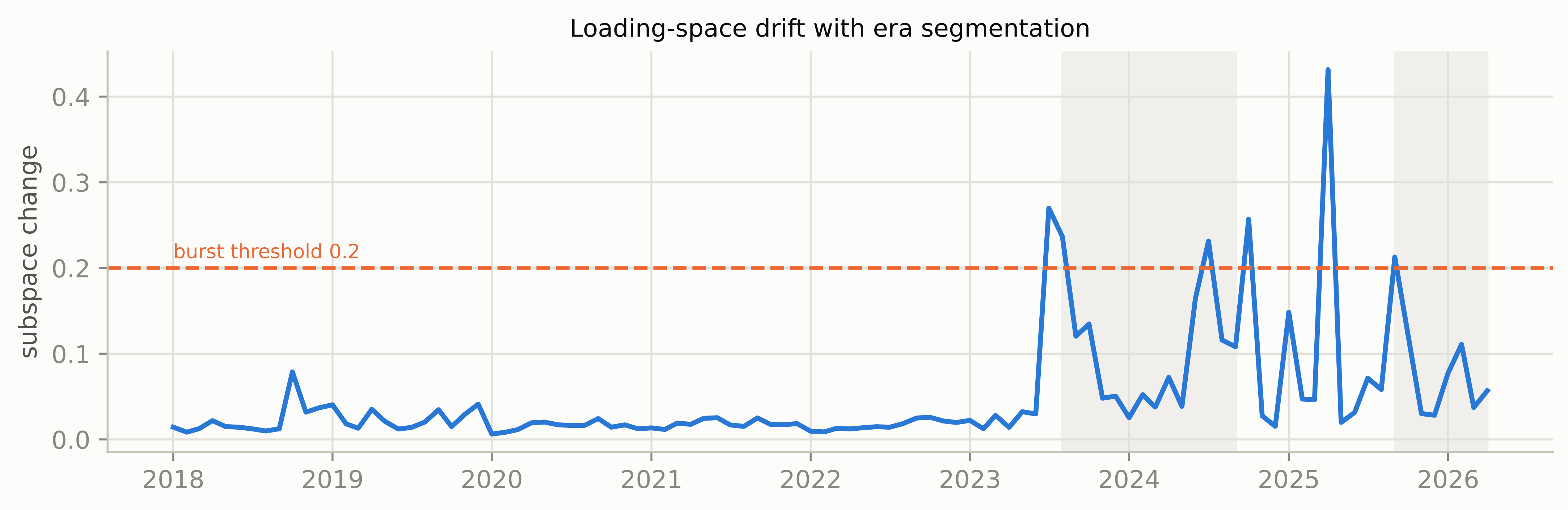
Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); pooled 2024 monthly MFM); outer columns rescaled to actual totals. CHK = China+HK. Data: Comtrade+TDM monthly panel.

2. The time-varying factor model

Era-anchored TV-MFM on the audited monthly panel (12m trailing window, k=4, $B levels, NNF basis), 2017-12..2026-04. Shaded bands = eras between structural breaks. Generated by scripts/tvmfm_monthly_anchored.py; summaries in results/mfm/tvmfm/.

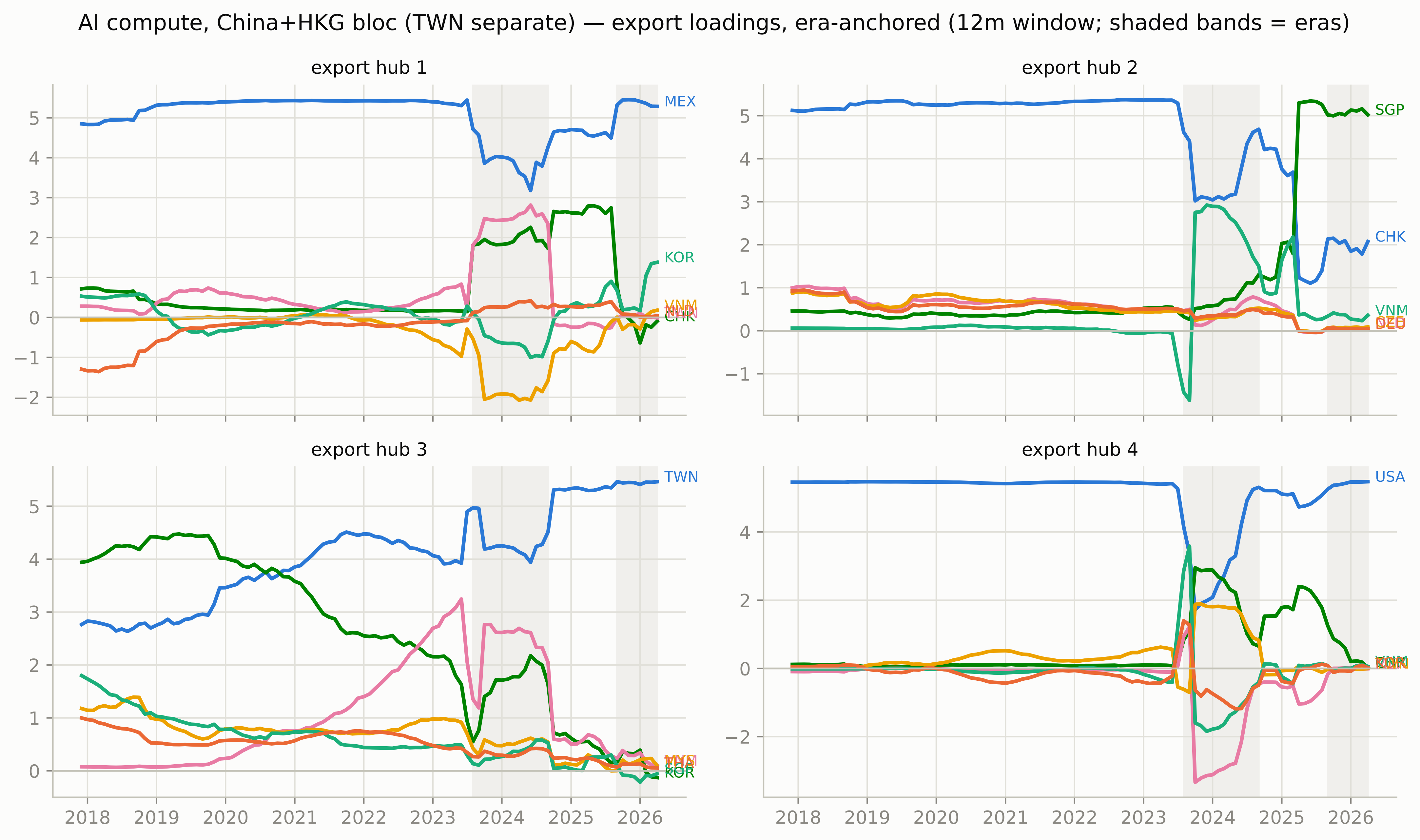
AI compute, CHK basis -- loading drift and eras

The preferred basis (entrepot churn netted; Taiwan separate). One unbroken 68-month era from 2017-12, then breaks at 2023-08, 2024-10, 2025-09.



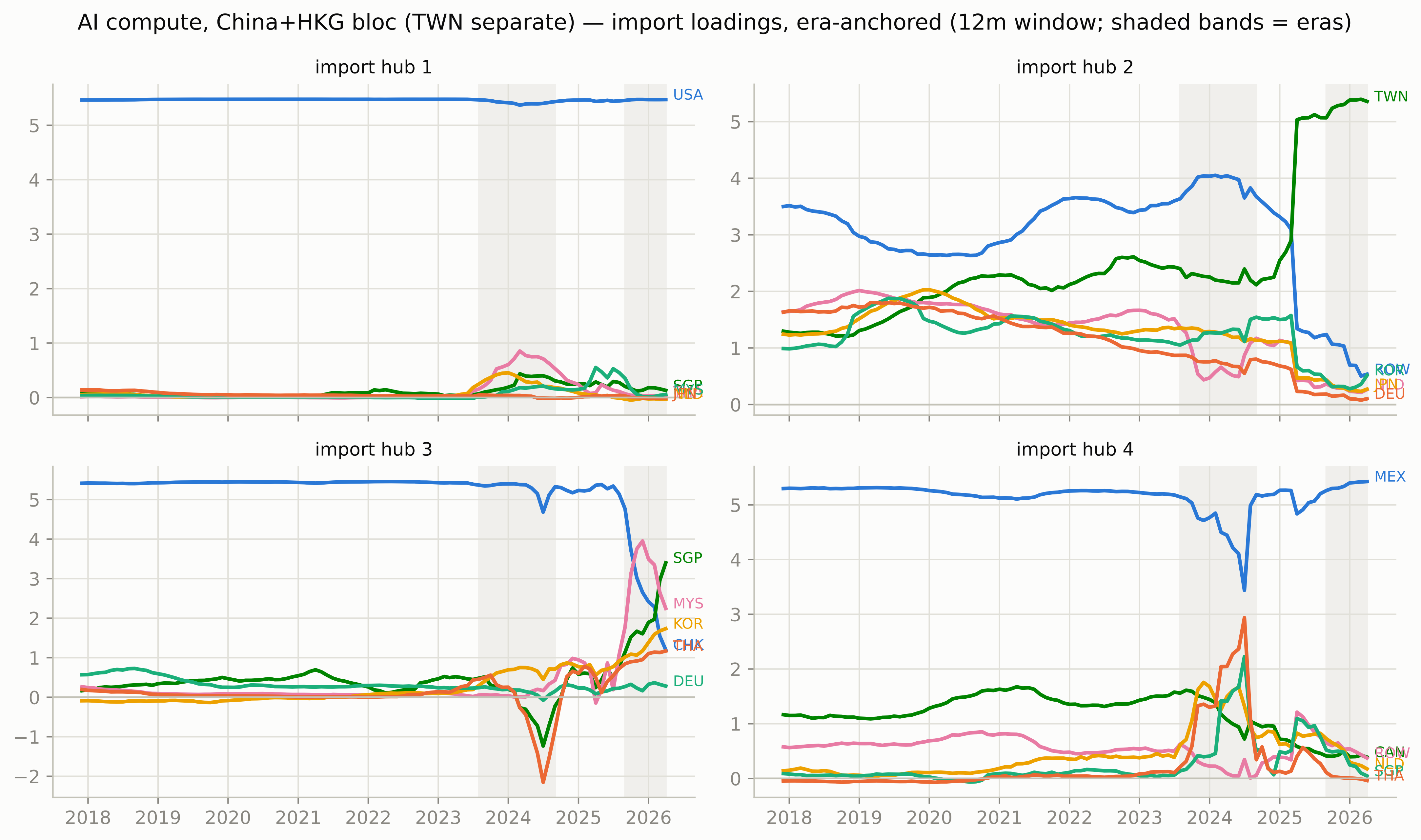
AI compute, CHK basis -- export-hub loadings

Per-hub country loadings through time. At 2024-10 the China-bloc hub loses its separate identity (cross-era similarity 0.45) and a Singapore-led hub appears.



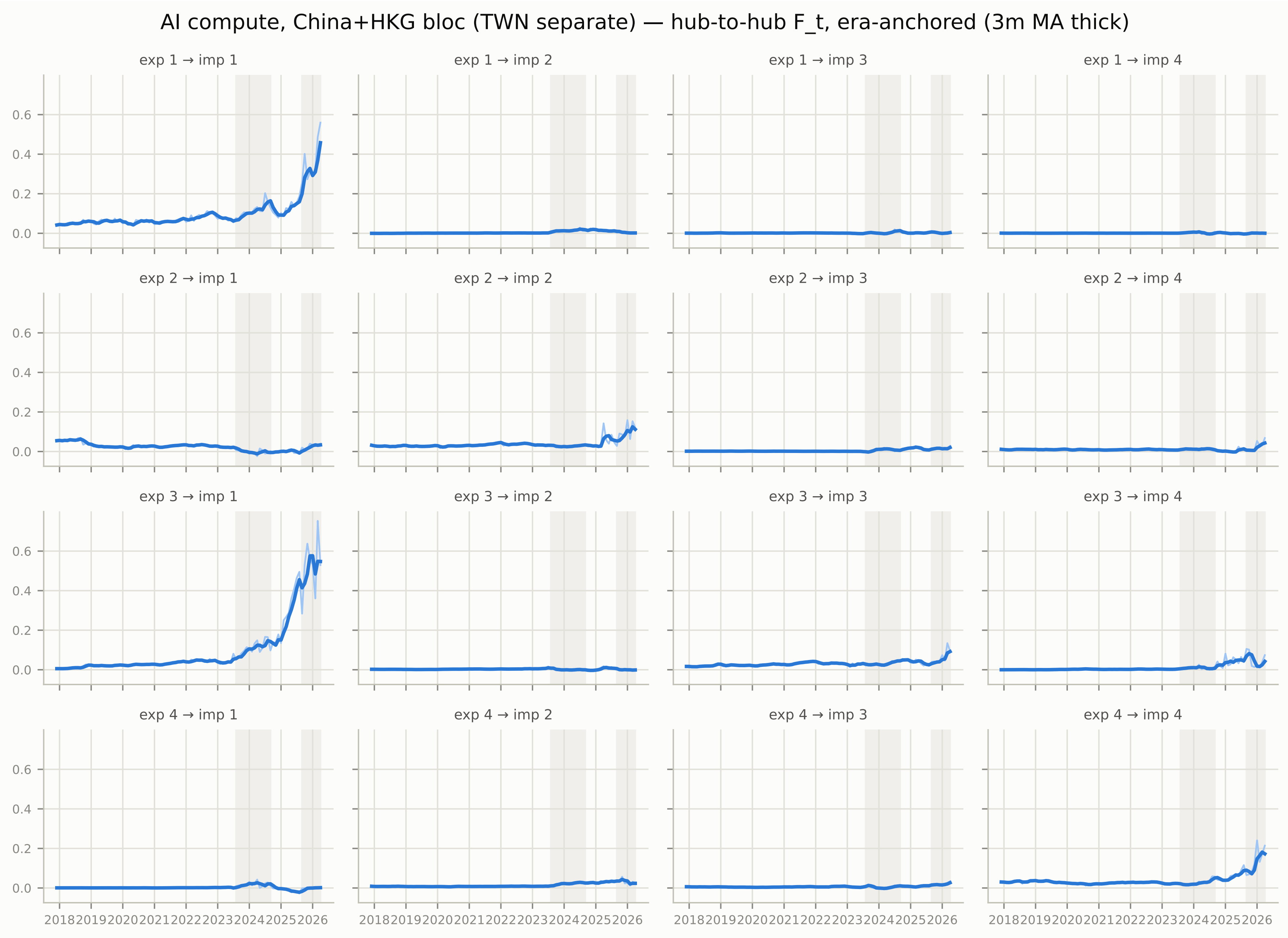
AI compute, CHK basis -- import-hub loadings

Import hubs are the stable side of the network (USA, Mexico, Taiwan, rest-of-world led).



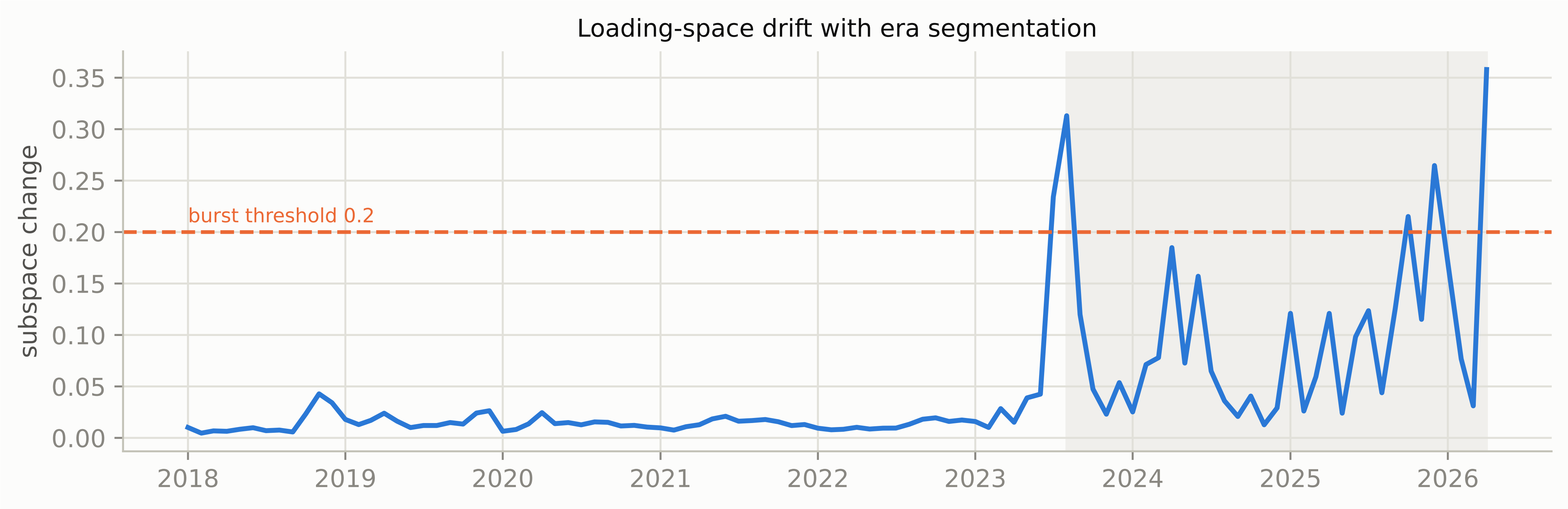
AI compute, CHK basis -- hub-to-hub intensities F_t

Each panel is one export-hub x import-hub channel in $B (3m MA thick).



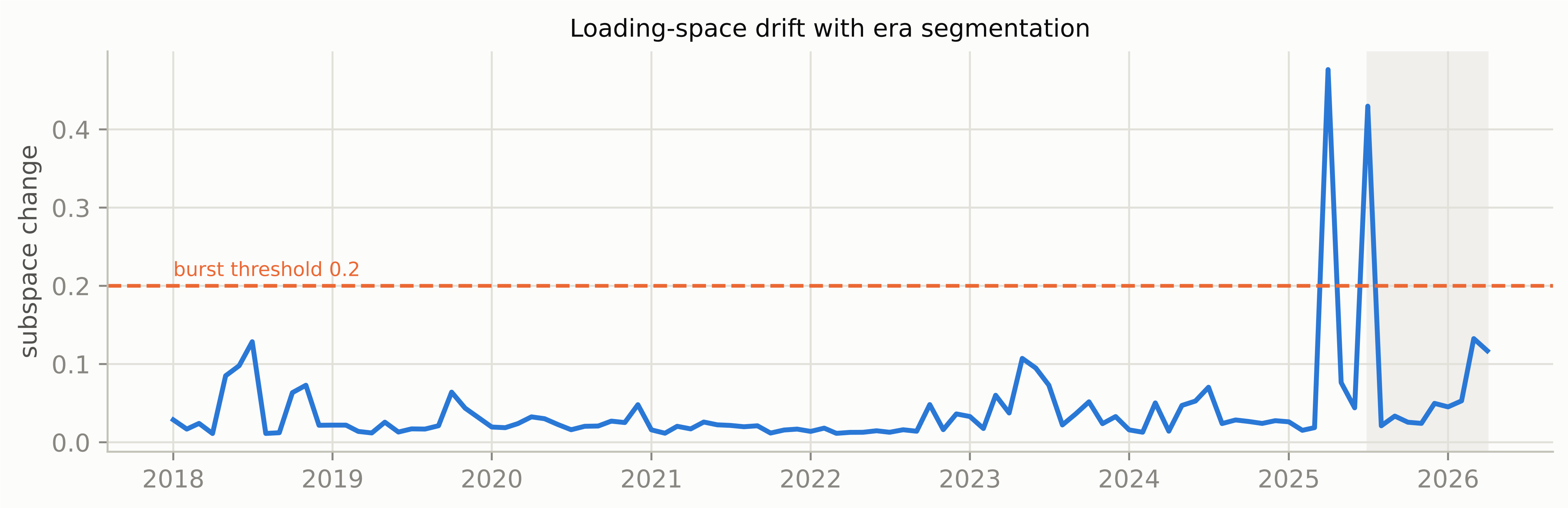
AI compute, country level -- drift

Without bloc merging the sample splits once: 2023-08. The demand event is the only dominant country-level break in nine years.



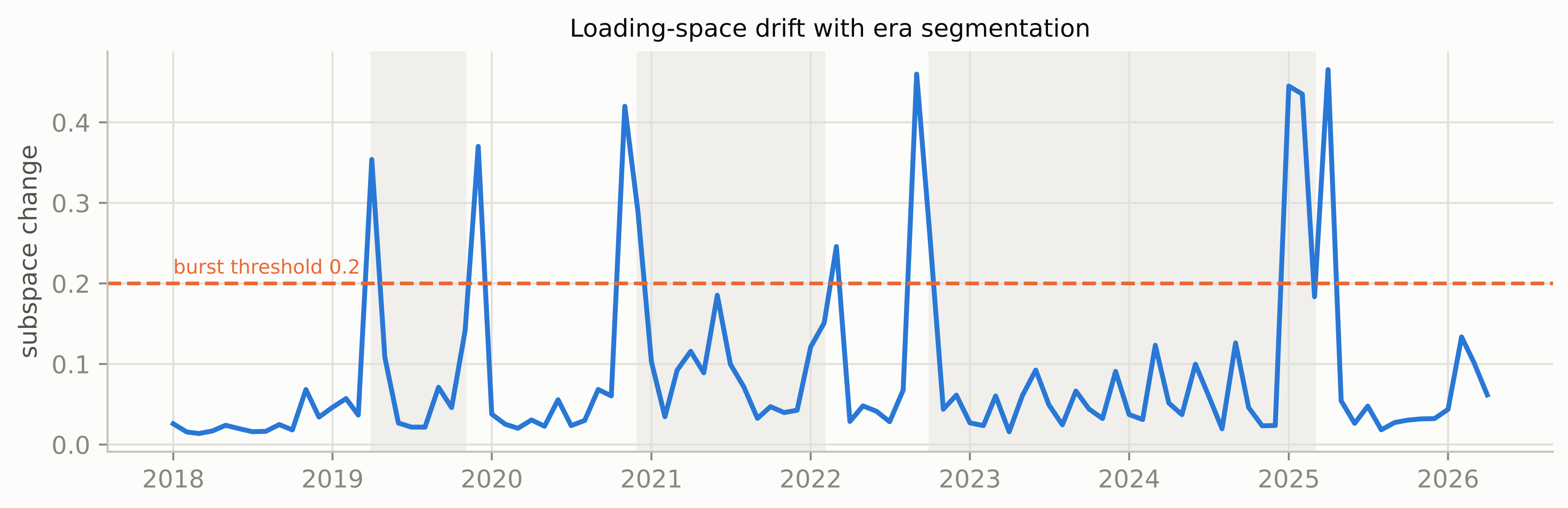
Parts (847330), country level -- drift

91 months without a structural break; the parts layer is the stable substrate of the network.



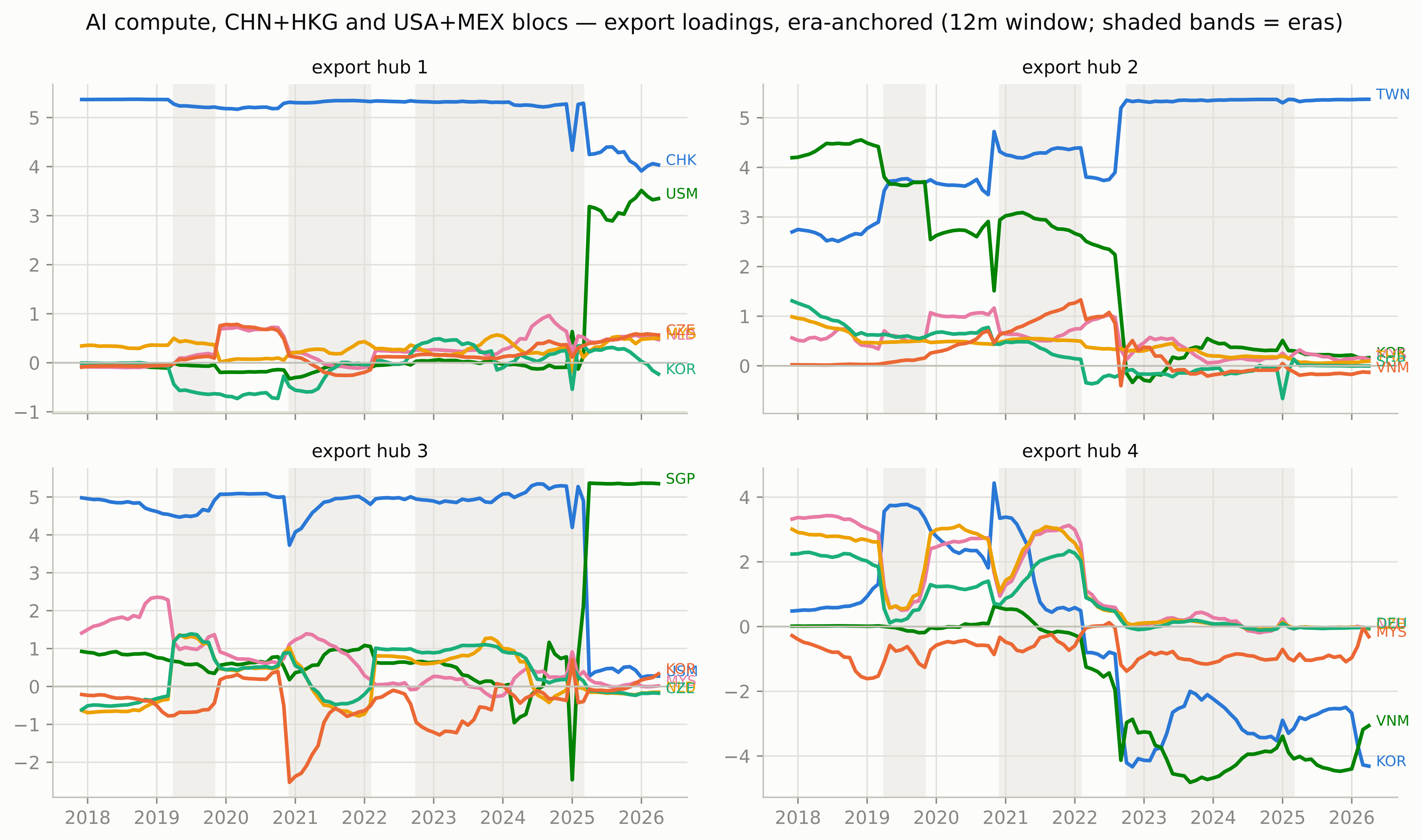
AI compute, double-bloc basis -- drift

CHK and USA+MEX both merged, isolating cross-bloc structure. Breaks at 2019-04/2019-12 (first tariff war), 2022-10 (first export controls), 2025-04 (tariffs): policy breaks blocs.



AI compute, double-bloc basis -- export-hub loadings

In the 2025-04 era the two blocs load together on one factor (CHK +3.97, USM +3.41) -- the entanglement signature unique to the tariff break.

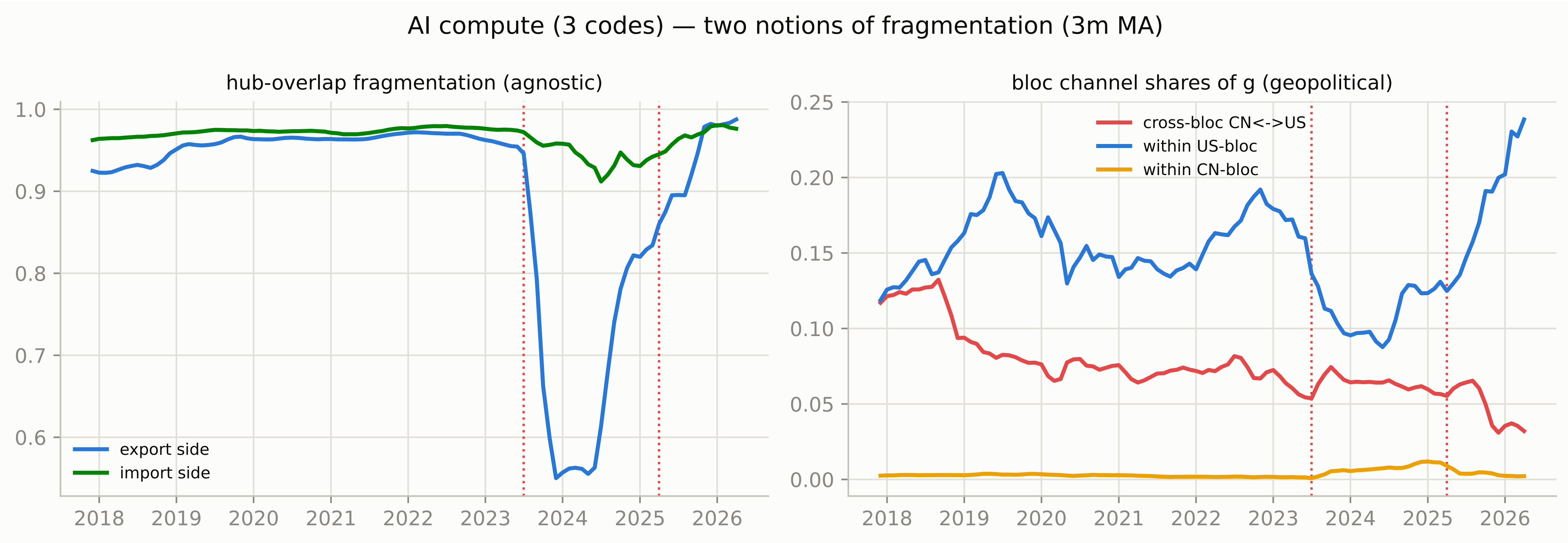


3. Concentration & fragmentation

Measure system of docs/concentration-fragmentation-nnf-mfm-trade.md computed monthly from the TV-MFM (CHK basis). Dotted lines mark 2023-07 and 2025-04. Interpretation: results/network_stats/findings.md. Generated by scripts/network_stats.py.

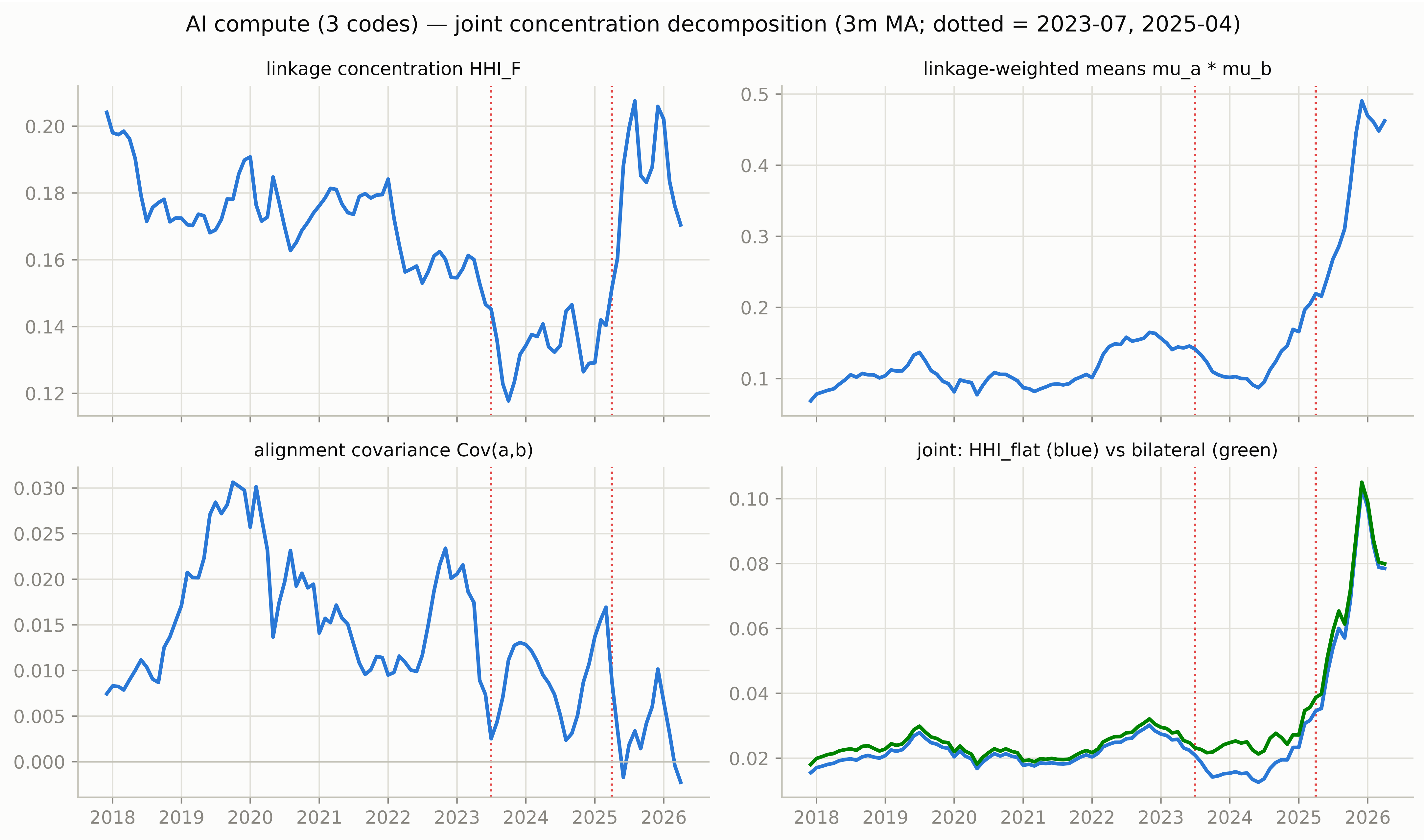
Two notions of fragmentation -- AI compute

Left: hub-overlap fragmentation (range-compressed under the anchored basis; mainly a diagnostic). Right: bloc channel shares -- the cross-bloc CN-US share falls ~70% over the sample in two legs (2019, then 2023 onward) with a plateau between.



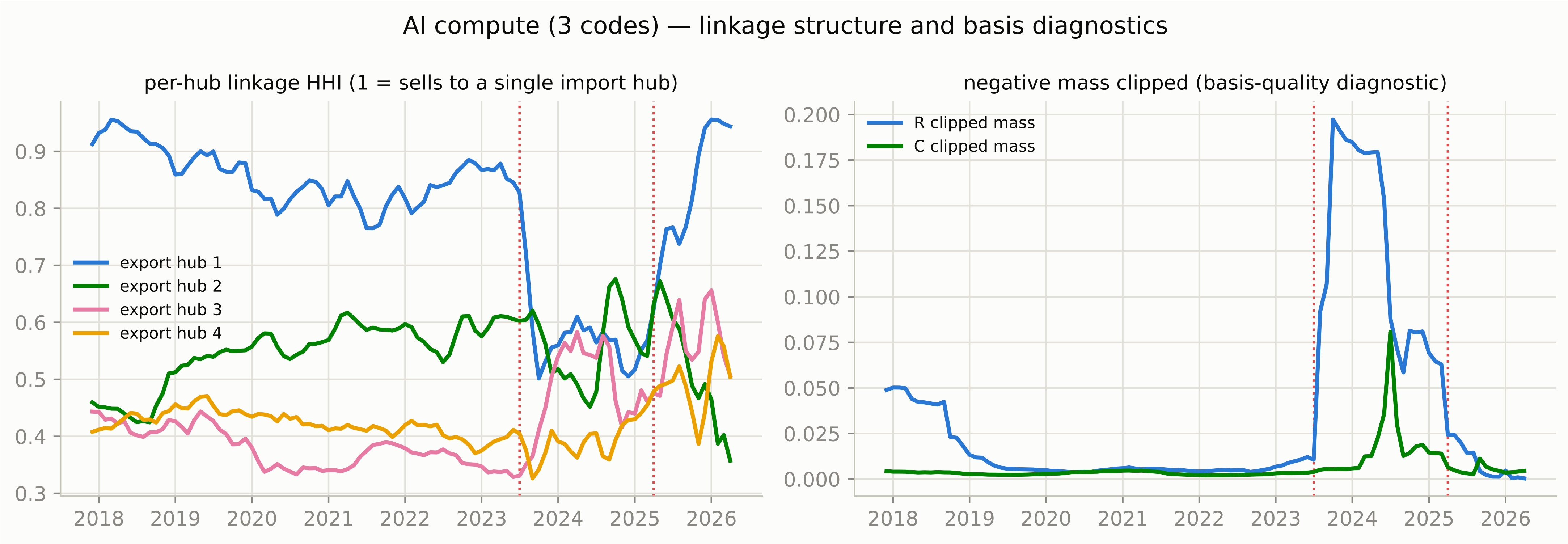
Joint concentration decomposition -- AI compute

HHI_flat = HHI_F x (mu_a mu_b + alignment covariance), exact identity. Alignment covariance is mildly positive from 2017, peaks ~2019, decays through the AI era: no compounding-concentration signature.



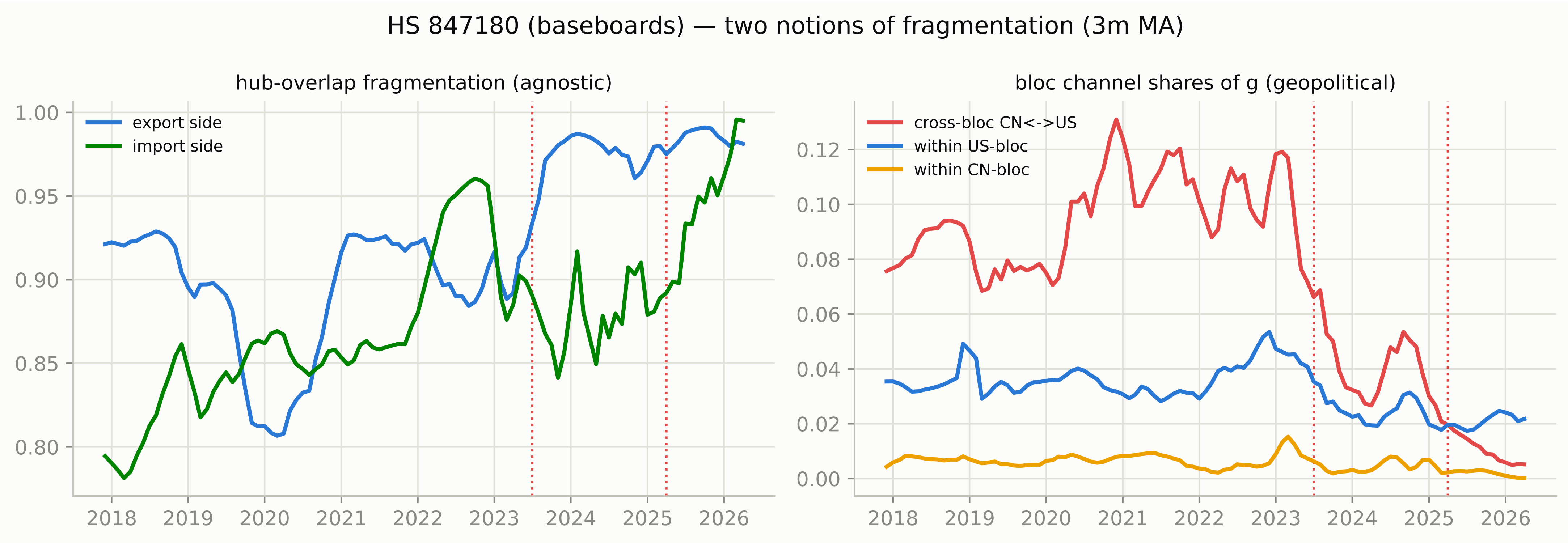
Per-hub linkage concentration and basis diagnostic -- AI compute

Left: does an export hub sell into one import program or several. Right: clipped negative mass (basis quality), elevated during the 2023-24 regime transition.



Fragmentation -- baseboards (847180)

The sharpest decoupling of the three codes: the cross-bloc share collapses ~20x from its 2021 peak while linkage concentration HHI_F rises 0.09 -> 0.48.



Fragmentation -- parts (847330)

Cross-bloc share halves but remains the highest of the three codes: the parts layer stays the shared substrate.